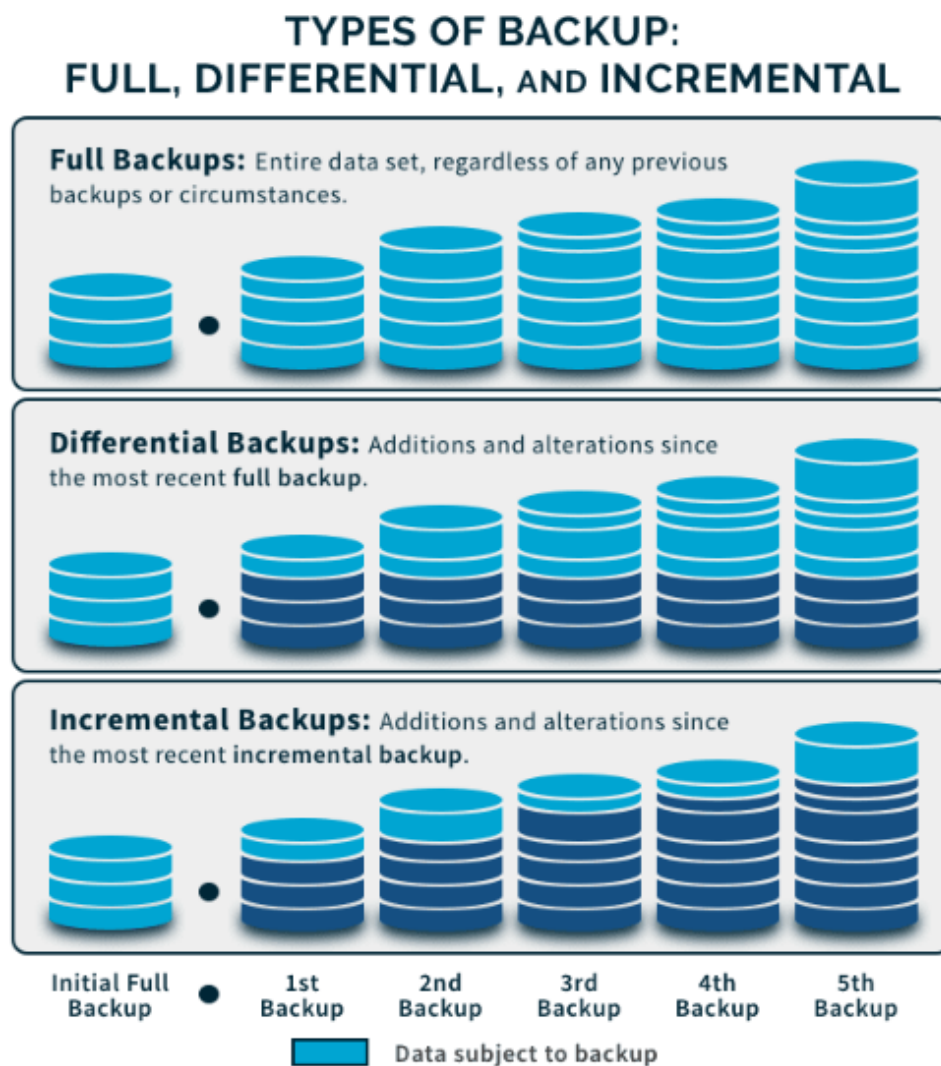


Types of Backup-58

1. Introduction

Backup methods differ in how data is copied and stored. Choosing the right type of backup depends on data size, storage availability, recovery needs, and how often changes occur. The goal is to balance storage cost, speed of backup, and recovery efficiency.



2. Types of Backup

1. Full Backup

- **Definition:** A complete copy of all selected data.
- **How it works:** Every file and folder is duplicated each time a backup is performed.

- **Advantages:**
 - Simple to manage.
 - Easiest and fastest restoration process since all files are stored in one set.
- **Disadvantages:**
 - Requires the most storage space.
 - Takes longer to complete compared to other methods.
- **Use Case:** Organizations typically perform a full backup weekly, followed by incremental or differential backups on other days.

2. Incremental Backup

- **Definition:** Only backs up data that has changed since the last backup (full or incremental).
- **How it works:** If a full backup was taken on Sunday, then Monday's incremental will copy only changes made since Sunday, Tuesday's will copy changes since Monday, and so on.
- **Advantages:**
 - Requires less storage space.
 - Backup process is fast.
- **Disadvantages:**
 - Restoring is slower, as it requires the last full backup plus every incremental backup taken after.
- **Use Case:** Daily backups where speed and storage efficiency are more important than fast restoration.

3. Differential Backup

- **Definition:** Backs up all data that has changed since the last full backup.
- **How it works:** If a full backup was taken on Sunday, Monday's differential backs up all changes since Sunday. Tuesday's backup includes Monday's changes plus Tuesday's new changes.
- **Advantages:**

- Faster restoration than incremental, since only the full backup and the last differential are required.
- **Disadvantages:**
 - Backup size grows each day until the next full backup is performed.
- **Use Case:** Balanced approach when quick recovery is important but storage resources are limited.

- After you select a backup method, you need to pick the most appropriate **rotation scheme** for your organization and network needs, which can reduce media costs and extend the longevity of your tapes while ensuring that every file is protected.

Verification

- Once data has been backed up it must be checked to ensure the backup was successful.

Each backup type has strengths and weaknesses. Full backups are reliable but resource-heavy. Incremental backups save space but slow recovery. Differential backups strike a balance between speed and efficiency. Mirror backups ensure real-time copies, while CDP provides the highest level of protection at higher cost. A strong backup strategy often combines multiple methods to achieve the best balance between cost, efficiency, and recovery speed.